

Distribution and Characteristics of Spiny Lobster (*Panulirus* spp.) Puerulus Collection Sites in Palawan, the Philippines

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Identifying the collection sites for the wild seeds of a highly valued species is crucial for the conservation and management of its wild population and habitat. This study is the first to document the distribution and the characteristics of spiny lobster puerulus collection sites in Palawan, Philippines. The data were obtained from various key informants, site visits, and historical imagery from Google Earth Pro and Sentinel 2A of Copernicus between January 2021–December 2023. The data were loaded into the QGIS v3.28 software to produce a map of each locality projecting the distribution and characteristics of spiny lobster puerulus collection sites. The results showed that the collection began in the municipality of Sofronio Española in 2019 and eventually spread southward and northward along the east coast of Palawan. The municipalities on the east coast of Palawan such as Sofronio Española, Narra, and Roxas had the highest number of collection sites, whereas fewer on the west coast such as in Rizal. Most of the collection sites were located generally along the intertidal zones or embayment with sandy-muddy to rubble substrates adjacent to mangrove forests, or within seagrass or seaweed, and coral reef ecosystems. To conserve and manage the spiny lobster puerulus fishery, the revision of the provincial ordinance on spiny lobster puerulus collection in Palawan should include the proper zoning of collection sites, imposing catch limits, and annual catch monitoring. Venturing into spiny lobster aquaculture can help reduce the fishing pressure and enhance the earnings of the people.

Keywords: distribution, habitat characterization, Palinuridae, remote sensing, spread

INTRODUCTION

The high market demand and price of adult spiny lobster have encouraged many small-scale fishers in various countries to engage in aquaculture (Jones 2010; Priyambodo *et al.* 2017; Macusi *et al.* 2019) utilizing the wild seeds (*e.g.* puerulus and juvenile) collected in coastal areas (Jones 2015; Priyambodo *et al.* 2017; Macusi *et al.* 2019; Mecha *et al.* 2022). Seedstocks of spiny lobster aquaculture are solely collected from the wild (Priyambodo and Jaya 2009; Jones 2010, 2018; Jones *et al.* 2019) due to the absence of commercial hatchery facilities and difficulties in hatchery

propagation (Barnard *et al.* 2011). This makes the collection of spiny lobster puerulus popular in many coastal areas and considered a lucrative source of income for many coastal inhabitants (Macusi *et al.* 2019; Priyambodo *et al.* 2020; Mecha *et al.* 2022, 2023).

Spiny lobster puerulus is a nektonic post-larval phase that comes after the oceanic planktonic larval phase known as phyllosoma larvae (Jeffs *et al.* 2005). Mostly, the phyllosoma of spiny lobster lasted up to 4–18 mo in some species involving 7–20 stages (Phillips and Sastry 1980; Jeffs *et al.* 2005) while drifting by ocean current (Bruce *et al.* 2000) away from continental shelves up to 1,500 km from the coast (Booth *et al.* 1998; Ehrhardt and Fitchett

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2010). After several months as a free-swimming larva, the phyllosoma will then molt as a puerulus which swims or drifts by the current back to the coast to find suitable habitat (Jeffs *et al.* 2005; Phillips and McWilliams 2009).

The ability of spiny lobster puerulus to reach the coast is strongly influenced by various factors (Thuy and Ngoc 2004; Jeffs *et al.* 2005; Briones-Fourzán and Lozano-Álvarez 2013; Priyambodo *et al.* 2017; Radhakrishnan *et al.* 2019; Priyambodo *et al.* 2020). Among these are the water movement (*e.g.* waves and current) due to seasonal changes, lunar phase, and the characteristics of the seabed (*e.g.* presence of crevices or holes in the reefs, algal flat, seagrasses, and mangroves) (Thuy and Ngoc 2004; Briones-Fourzán and Lozano-Álvarez 2013; Priyambodo *et al.* 2017; Valadez-Manzano *et al.* 2017; Jones *et al.* 2019; Radhakrishnan *et al.* 2019; Priyambodo *et al.* 2020). Upon reaching the coast, the pueruli of various spiny lobster species (*e.g.* *Panulirus argus*, *P. cygnus*, *P. interruptus*, *P. ornatus*, *P. japonicus*, and *P. gracilis*) exhibit ontogenetic habitat shift from the post-larval settlement habitat of macroalgae, kelp, or seagrass to benthic crevices sheltering aggregation of larger juveniles, sub-adults and adults (Briones-Fourzán and Lozano-Álvarez 2013; Radhakrishnan *et al.* 2019).

The above-mentioned factors influence both the abundance of puerulus settlement and collection activities (Booth and Phillips 1994; Briones-Fourzán *et al.* 2008; Dao and Jones 2015; Priyambodo *et al.* 2017, 2020). For example, in the study of Eggleston *et al.* (1998) – in Exuma Sound in the Bahamas – the location with the highest puerulus settlement had a regionally persistent offshore eddy that pushed water onshore. In Vietnam, the collection sites are on the east coast facing the South China Sea, where the seasonal wind brings more puerulus from open water to the coast (Dao and Jones 2015; Jones *et al.* 2019). In Indonesia, collection sites with abundant puerulus are located on the south and southeast coasts facing the Indian Ocean (Priyambodo *et al.* 2020). Some of these coastal areas are characterized by enclosed bays having medium to strong water current, the presence of freshwater intake from the rivers, strong to very strong wind with 2–5 m wave height, sandy-muddy substrate, and pristine coastal water (Priyambodo *et al.* 2020). In the Philippines, the collection sites within Balete Bay, Davao Oriental have wide sand or sandy-muddy and coralline flats substrate, and a water depth of 1–6 m (Macusi *et al.* 2019). Such information on the collection sites is crucial for the conservation and management of the spiny lobster puerulus population and the surrounding environments (Jones *et al.* 2019; Macusi *et al.* 2019; Priyambodo *et al.* 2020). In Palawan, Philippines, spiny lobster puerulus collection activities only started in 2019 and it became a lucrative source of income for many Palawenos during the

COVID-19 pandemic [see Mecha *et al.* (2022)]. However, information about the characteristics of the collection sites is still lacking which is an important step to craft local policies for a sustainable puerulus fishery and related industry in Palawan. Hence, this study was conducted to document the distribution and the characteristics of coastal areas where puerulus collection has been reported.

MATERIALS AND METHODS

Study Sites

The study covered eight coastal municipalities and one city in Palawan reported to have an ongoing collection of spiny lobster puerulus [see Mecha *et al.* (2022)]. These include the municipalities of Taytay, Dumarán, Roxas, Aborlan, Narra, Sofronio Española, Brooke's Point, Rizal, and the City of Puerto Princesa (PPC). Except for Rizal located on the west coast (facing the West Philippine Sea) and Aborlan, PPC, and Taytay with west and east coast areas, the other municipalities are situated on the east coast of Palawan facing the Sulu Sea (Figure 1). These seas are known as the apex of marine biodiversity – the Sulu-Sulawesi Marine Ecoregion (SSME), which serves as an important fishing ground in the region (ADB 2014).

Distribution of Spiny Lobster Puerulus Collection Sites

The spiny lobster puerulus collection sites were identified by 30 key informants (KIs) [puerulus collectors (20) and government employees (10)] who responded to the social media post (Albarico *et al.* 2021) from January 2021–December 2023 and personal site visits in selected localities (Figure 1). All KIs were also asked about the year when puerulus collection started in each locality. In addition, the Google Earth Pro (GEP) images of all coastal localities with confirmed spiny lobster puerulus collection from 2019–2023 were also checked for historical puerulus collection activities. However, GEP has no images in 2023; hence, the satellite validation only covers 2019–2022. In addition, some of the coastal localities have no clearer GEP images; hence, only those images with recognizable puerulus settlement traps (see Figure 2) were included in this study. These images were georeferenced in QGIS v.3.28 software to map out the estimated position and the extent of puerulus settlement traps installed within the coastal area. Each clear image was also assessed to differentiate it from other coastal structures such as seaweed farms or fish cages (Figure 2).

Characterization of Collection Sites

The coastal habitats (*e.g.* seagrass beds, coral reefs, and bottom substrates) in each locality with collection

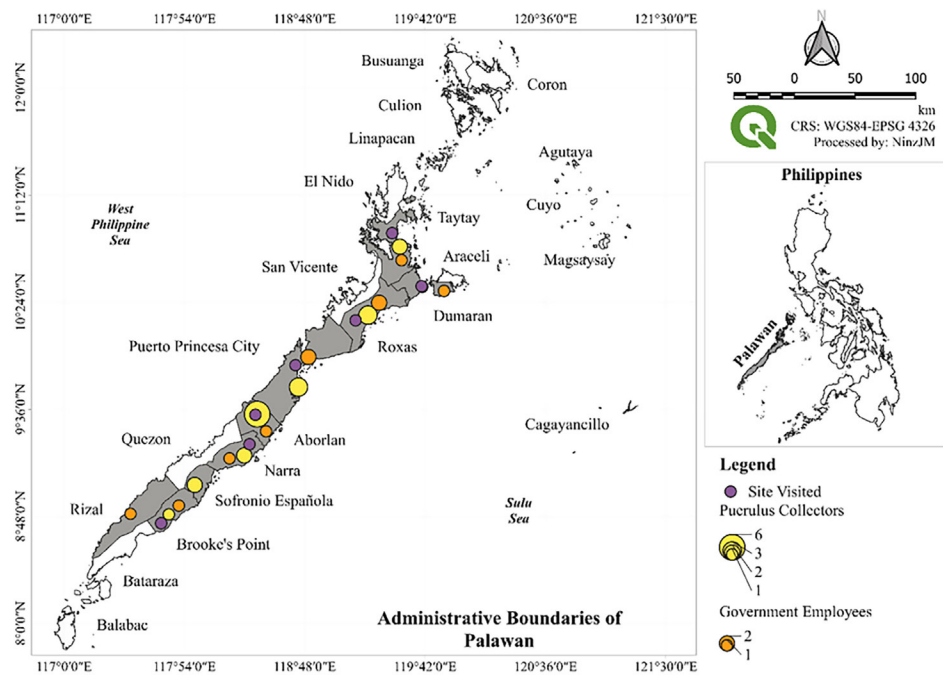


Figure 1. Map of Palawan in the west side of the Philippines, the nine localities (eight municipalities and one city) where people engage in spiny lobster puerulus collection (gray layer), and the number of key informants (KIs) and sites visited.



Figure 2. Google Earth Pro imagery sensed in 2020 shows the differences between the spiny lobster puerulus settlement traps in Brooke's Point (above) and the seaweed farm in Taytay, Palawan (below).

activities were determined and characterized using the 2020 Sentinel 2A images downloaded on the Copernicus website. The Sentinel 2A images were resampled and analyzed in SNAP (Sentinel Application Platform) v9 software to obtain the class indices of all the benthic habitats. The analyzed images were then further cleaned using the QGIS v3.28 to separate, classify, and generate a layer of each benthic habitat. Validation of all coastal environs within the collection sites was done by interviewing some puerulus collectors, using Coral Allen Atlas (www.allencoralatlas.org), and visiting the sites. All generated layers of satellite images per municipality including the puerulus settlement traps were combined and saved as a single image.

RESULTS

Distribution of Spiny Lobster Puerulus Collection Sites in Palawan

As claimed by some KIs and based on the historical images from GEP, the collection of spiny lobster puerulus in Palawan began in Sofronio Española in 2019 (Figure 3). During the succeeding years, the collection activities spread toward southern Palawan – particularly, in Brooke's Point and Rizal, as well as on the west and east coast of Taytay north of Palawan. In 2021, the puerulus collection spread mostly throughout the central-eastern portion of Palawan, where the high concentrations of collection sites were found along the coasts of Dumarán, Roxas, PPC, Aborlan, Narra, Sofronio Española, and Brooke's Point.

The majority of these collection sites were located on the east coast of Palawan with fewer on the west coasts of Taytay, Aborlan, and Rizal. These areas remained as the collection sites until 2022 and 2023 (Figure 3).

Characteristics of Collection Sites

Most of the spiny lobster puerulus collection sites were located within the intertidal zone or embayment with nearby mangroves, seagrasses, and coral reefs (Figure 4). For example, the high concentration of collection sites in Roxas, Narra, Sofronio Española, and Brooke's Point were found within the embayment with sparse coastal environments. Meanwhile, some localities with few collection sites such as Taytay, Dumarán, Aborlan, and Rizal were observed in intertidal zones with dense coastal environs (Figure 4).

Looking at the specific areas in localities with high concentrations of collection sites like *Barangay* I, II, and IV, Roxas, Palawan, the collection sites were found along the shallow embayment with sandy-muddy-rubble substrate and coral, seagrass, or seaweed habitat. In Punang, Sofronio Española, the collection sites were situated at the midsection of the cove with the presence of river outflow and mixed substrate extended toward the opening of the coast with nearby mangroves, seagrass or seaweeds, and coral reefs. Meanwhile, the collection sites in Barong-Barong and Ipilan, Brooke's Point were found on a sandy beach with the presence of river outflow, sparse corals, and seagrass (Figure 5).

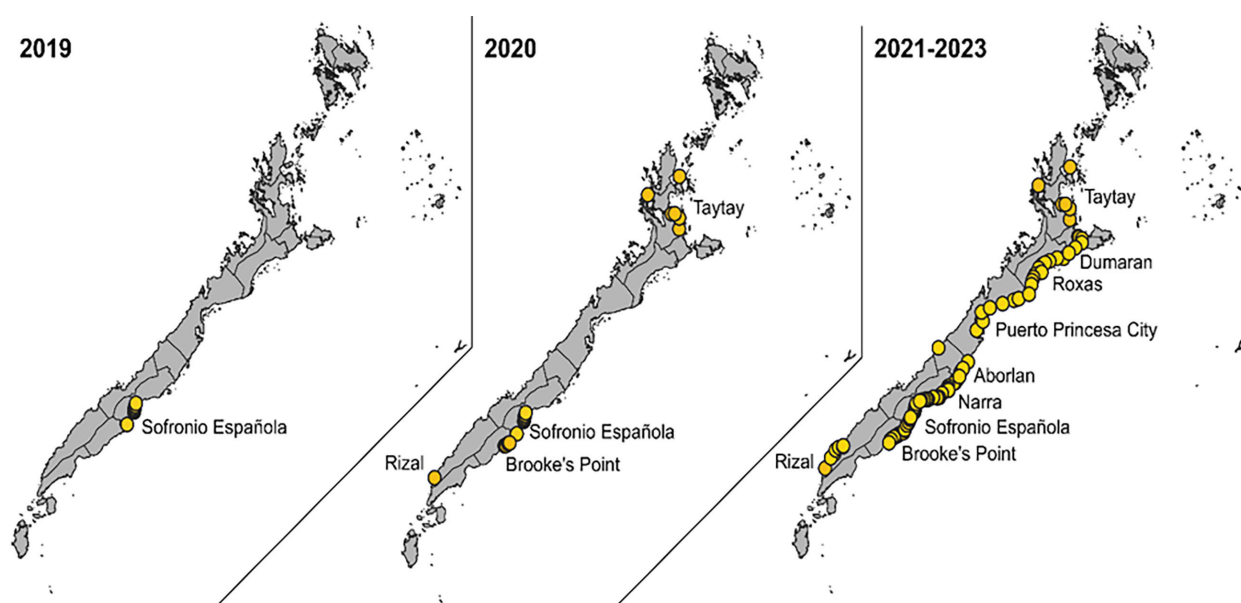


Figure 3. The distribution and spread of the spiny lobster puerulus collection sites from 2019–2023 in the province of Palawan, the Philippines.

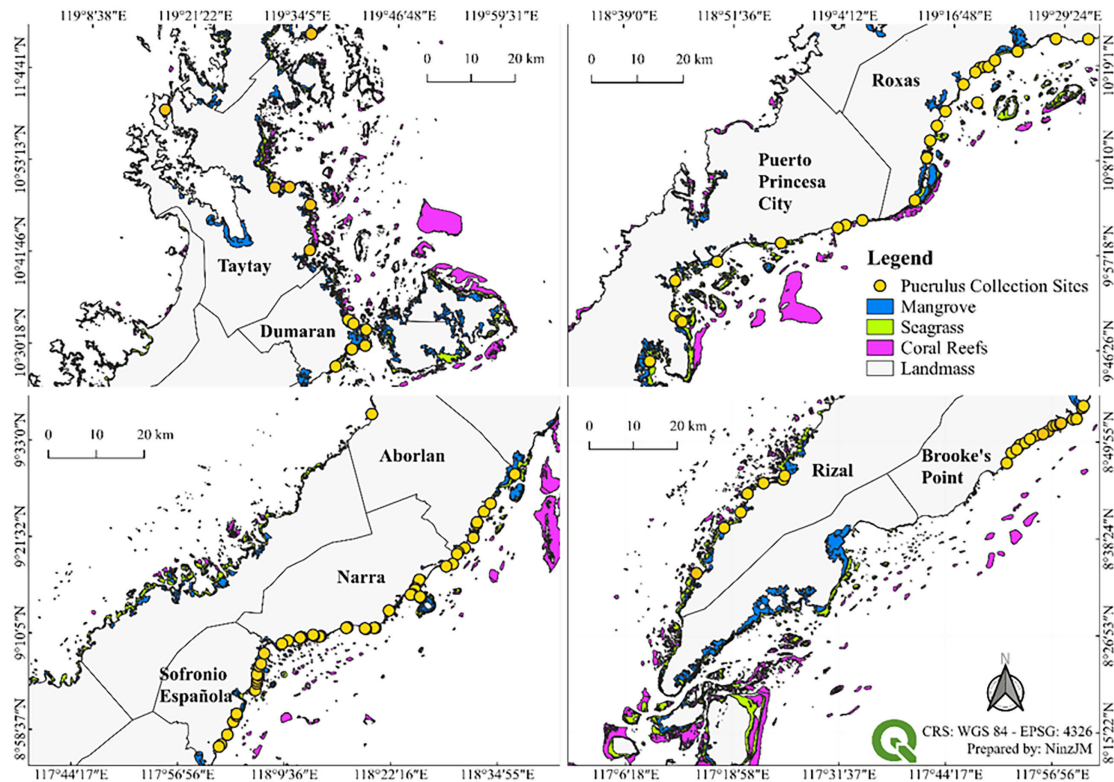


Figure 4. The characteristics of spiny lobster puerulus collection sites in Palawan, the Philippines.

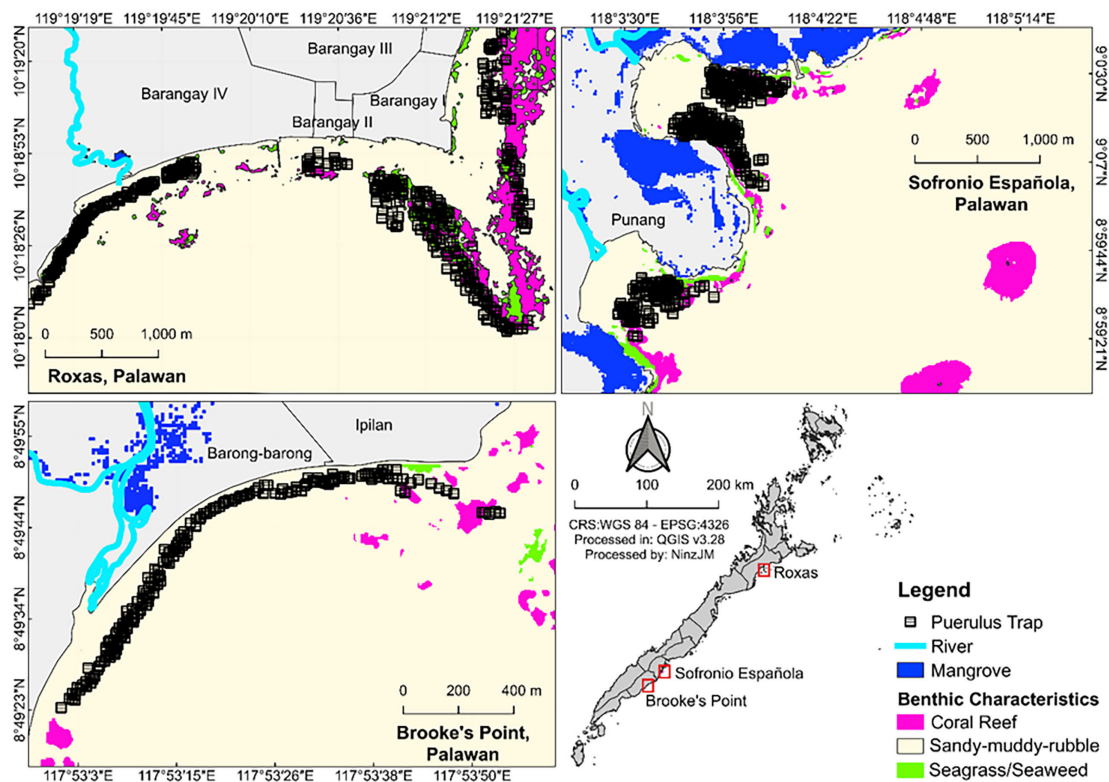


Figure 5. The characteristics of the coastal area within the selected collection sites in Roxas, Brooke's Point, and Sofronio Española, Palawan.

DISCUSSION

Distribution of Spiny Lobster Puerulus Collection Sites

According to some KIs, the start of the spiny lobster puerulus collection in Sofronio Española in 2019 was due to an individual Vietnamese buyer who introduced the methods of collection [see Mecha *et al.* (2022)]. Vietnam is the first country in the world to establish significant spiny lobster aquaculture using the wild puerulus and juvenile, which started in 1995 (Jones 2010; Jones *et al.* 2019). Eventually, while the demand for farmed adult spiny lobster increases, the supply of wild seeds in Vietnam becomes unstable due to the increase of lobster farmers (Jones 2010) and the seasonality of puerulus settlement (Priyambodo *et al.* 2020). This explains the introduction of collection methods by some Vietnamese individuals to other neighboring countries like the Philippines to supply the increasing demand for puerulus in Vietnam (Macusi *et al.* 2019; NFRDI 2020; Mecha *et al.* 2022). Priyambodo *et al.* (2020) also mentioned that some lobster fishers in Indonesia illegally exported puerulus to Vietnam due to the high buying price. On the other hand, the spread of collection in Palawan from 2019–2023 is due to the high price of puerulus that caught the interest of many coastal inhabitants, which became an alternative source of livelihood during the COVID-19 pandemic period. For example, a single puerulus can be sold from PHP 150 (USD 2.70) to PHP 200 (USD 3.60) for the crystal and PHP 80 (USD 1.44) to PHP 150 (USD 2.70) for pigmented [see Mecha *et al.* (2022, 2023)].

Meanwhile, the distribution of collection sites concentrated on the east coast of Palawan is similar to Indonesia, where puerulus is abundant only in the south and southeast coastal islands (Priyambodo *et al.* 2020). According to Thuy and Ngoc (2004), Phillips and McWilliams (2009), Dao *et al.* (2015), and Priyambodo *et al.* (2020) coastal waves and the sea surface currents are the main driving force of high puerulus settlement since they bring back the late-stage phyllosoma near the coast. The east coast of Palawan facing the Sulu Sea is also exposed to internal waves [see Earth Observatory (2003)] and sea surface current mostly comes from various coastal areas that are part of the SSME known as the apex of marine biodiversity (ADB 2014; He *et al.* 2017), including the Tubbataha Reefs Natural Park. As shown in Figure 6A, the two-year sea surface current within the Sulu Sea during the peak season of collection in Mar [see Mecha *et al.* (2023)] continuously flows toward the eastern coast of Palawan possibly transporting a high number of puerulus. The estimated sea surface current in the Sulu Sea changes every August, which explains the lean season (Figure 6B) [see Mecha *et al.* (2023)]. In Indonesia, the abundance of puerulus in the southern and southeast coastlines is attributed to the sea surface current coming from the northern and southern hemispheres,

which have various sources of breeding grounds for adult spiny lobster (Dao *et al.* 2015; Priyambodo *et al.* 2020). Meanwhile, the westward direction of the sea surface current on the west coast of Palawan during the peak season could have hindered puerulus settlement except in Rizal – the southern tip of Palawan, where the current from the Sulu Sea still reaches this locality (Figure 6A). This low to absence of collection sites on the west coast is supported by the study of Villanoy (2004) and Dao *et al.* (2015), where the suggested current flow and dispersal of *Panulirus ornatus* puerulus positioned in northwest Palawan only reached the west coast of Palawan and the coastal water of Vietnam. Meanwhile, despite the different sources of sea surface currents in Palawan, the collection season [March–August; see Mecha *et al.* (2023)] is similar to Indonesia (Dao *et al.* 2015; Priyambodo *et al.* 2020) and in Balete Bay, Davao Oriental (Dao *et al.* 2015; Macusi *et al.* 2019) but different from Vietnam [October–March; see Dao *et al.* (2015)]. These similarities of puerulus collection seasons in Palawan, Davao Oriental, and Indonesia indicate synchronous spawning over a wide geographic region, particularly within the SSME. Although the seasonal variation of sea surface current in the adjacent seas of Palawan corroborates with the differences in the distribution of collection sites and seasonality of puerulus abundance (Figure 6), larval dispersal modeling studies in relation to the monsoon and spawning time are highly recommended. Identifying the origin of breeding grounds, and the timing of larval dispersal might be the most effective management measure to ensure a sustainable supply of puerulus in Palawan.

As of this date, only Palawan (Mecha *et al.* 2022, 2023) and Balete Bay in Davao Oriental (Macusi *et al.* 2019) have published information related to puerulus collection in the Philippines. While some areas such as Surigao and Zamboanga were reported to be the hotspot of puerulus, information related to the collection activities in the said areas is lacking (NFRDI 2020). Some coastal areas on the east coast of Palawan have no reports of collection in 2021, but while writing this paper, some of these areas became collection sites in 2022 and 2023. Hence, continuous monitoring in terms of collection site expansion and monthly catches in each of the localities can aid the local government in creating policies and ordinances affecting the management of the spiny lobster puerulus fishery and the immediate environs.

Characteristics of Collection Sites

The sandy-muddy to rubble substrate within intertidal zones or embayment with weak to moderate waves, and the presence of adjacent coastal ecosystems (mangroves, seagrass, corals) in collection sites in Palawan have resemblance in Vietnam (Thuy and Ngoc 2004), Indonesia (Tewfik *et al.* 2009; Priyambodo *et al.* 2020), and Balete

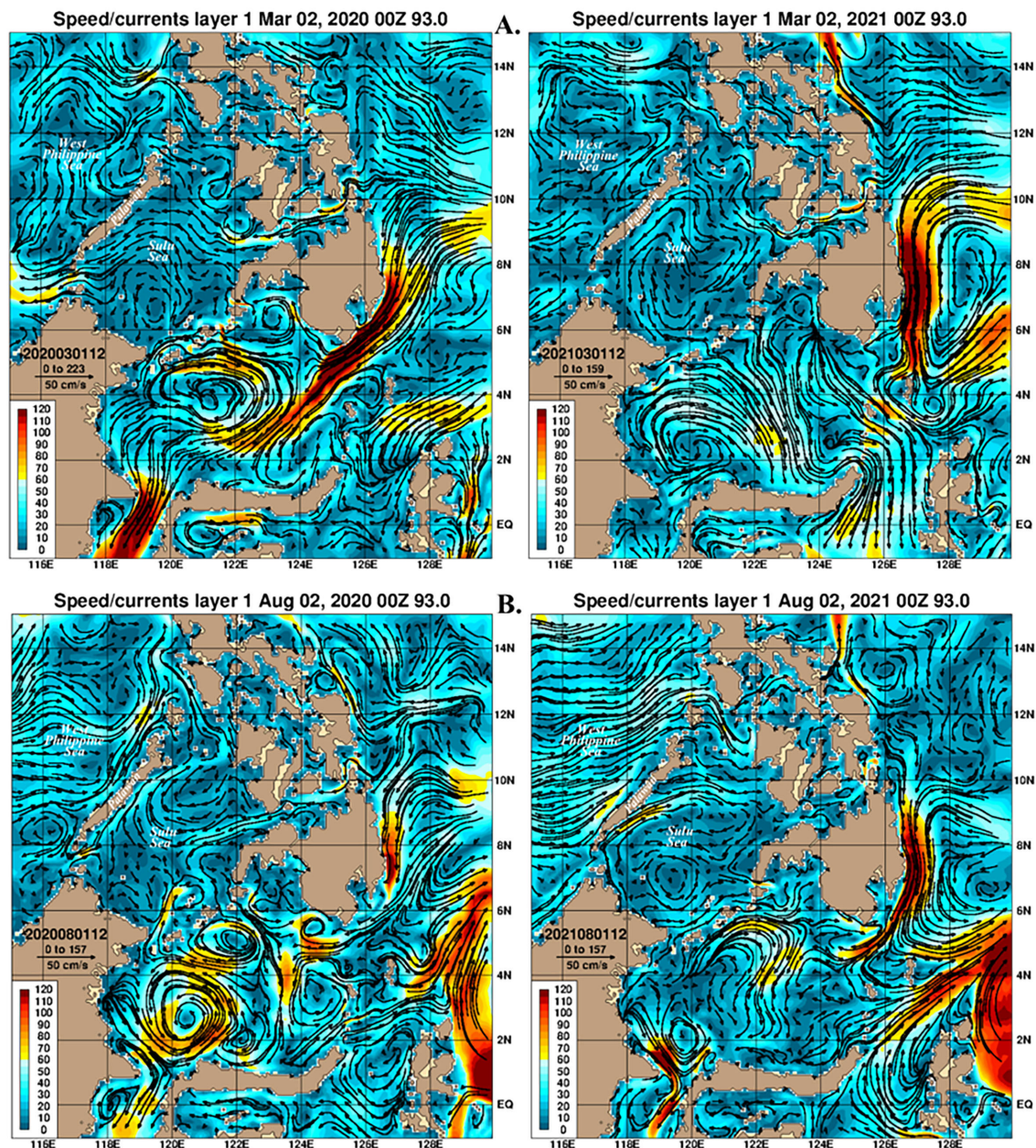


Figure 6. The estimated sea surface current of adjacent seas (Sulu Sea and West Philippine Sea) in Palawan during the start of puerulus collection season in March 2020 and March 2021 (A), as well as during the lean season of puerulus collection in August 2020 and August 2021 (B). Data extracted from Naval Research Laboratory HYCOM Consortium for Data Assimilative Ocean Modeling. (<https://www7320.nrlssc.navy.mil/GLBhycomce1-12/skill.html>)

Bay in Davao Oriental (Macusi *et al.* 2019). These aforementioned environmental conditions provide a safe refuge that attracts the late-stage phyllosoma (puerulus) along the coastal fringe to swim toward the preferred habitat (Jeffs and Holland 2000; Thuy and Ngoc 2004) rather than the water parameter conditions of the coastal area (Jeffs *et al.* 2005). However, according to Priyambodo *et al.* (2020), the presence of a coastal habitat favorable to puerulus metamorphosis and settlement is just a portion of

environmental conditions for a high abundance of puerulus. Another environmental condition that contributes to the settlement of puerulus on the coast is the turbid water due to the river outflow (Jones *et al.* 2019; Priyambodo *et al.* 2020), which was also observed in most of the collection sites in Palawan (see Figure 5). Despite this, puerulus caught near the river mouth exhibited high mortality rates as also observed in Indonesia (Priyambodo *et al.* 2020). In addition, spiny lobster puerulus is a non-burrowing

species (Radhakrishnan *et al.* 2019) and species such as the *P. ornatus* and *P. japonicus* prefer to settle in the hard-perforated substrate suitable to their body size surrounded by macroalgae or seagrasses (Yoshimura *et al.* 1994; Dennis *et al.* 1997). Other species prefer to settle directly in coastal vegetations such as the edges of mangroves, seagrass meadows, and macroalgal flats, which increase the provision of food resources and the amount of coverage for the early benthic stages (Beck *et al.* 2001; Briones-Fourzán *et al.* 2008). Therefore, identifying the dominant species at each collection site is crucial since some spiny lobster species exhibit different ontogenetic behavior in selecting suitable habitat (Briones-Fourzán and Lozano-Álvarez 2013; Radhakrishnan *et al.* 2019; Damora *et al.* 2021).

Conservation Management of Spiny Lobster and Surrounding Habitats

The popularity of spiny lobster puerulus collection in Palawan as a lucrative source of alternative livelihood during the COVID-19 pandemic (Mecha *et al.* 2022, 2023) showed a rapid expansion along the eastern coast, bringing potential threats to coastal marine ecosystems (*e.g.* seagrass beds and coral reefs). Additionally, the trap structures installed in coastal areas also interfere with the safety of navigation. Some collectors also utilized unsustainable collection methods such as the use of fine mesh nets combined with light at night as an attractant to catch spiny lobster puerulus. This type of collection method not only caught spiny lobster puerulus but also the fingerlings of other high-value fishes that attract to the light. Spiny lobster puerulus caught using this method are reported to suffer a low survival rate due to the stress made by dragging and lifting the net. Currently, there is a provincial ordinance implemented in Palawan for spiny lobster puerulus collection (Provincial Ordinance No. 2475, series of 2020), which is an adaptation from Fisheries Administrative Order No. 265 (BFAR 2020). However, this ordinance only covers the buying prices. Therefore, to effectively conserve the spiny lobster puerulus and protect the coastal habitats and other important marine species in Palawan, the provincial ordinance should also include the following: [1] designation of appropriate zones (*e.g.* in areas with fewer seagrass and coral reef habitats) with the use of sustainable collection methods; [2] imposing catch limit per collector; and [3] mandatory removal of settlement traps during the off-season (from September–December), including the proper disposal of nylon and polypropylene ropes as hanging lines and tying materials for settlement traps. It is important to identify and protect other puerulus settlement sites to ensure constant replenishment of the breeding populations. Imposing a catch limit can also protect the wild seedstocks from possible collapse, which could affect the replenishment of the adult population

in the reefs. In addition, zoning and mandatory removal of settlement traps during the off-season may also help reduce the possible risks that the installed settlement traps may cause to navigation, ecotourism, and other marine organisms (*e.g.* derelict fishing). Additionally, continuous monitoring of collection sites such as water depth, habitat or substrate characteristics, and terrestrial influences (*i.e.* erosion, outflow) could help further understand the dynamics of puerulus settlement. These could be used as an additional basis for policy recommendations and coastal management programs toward a sustainable puerulus fishery in Palawan. Moreover, government support in promoting spiny lobster aquaculture can help reduce the export of wild spiny lobster puerulus as what happened in Davao Oriental, the Philippines (Macusi *et al.* 2019). Although economic data on spiny lobster puerulus collection and trade in Palawan is lacking, it is believed that this fishery sector is a multi-billion-dollar industry in Vietnam and Indonesia (Petersen and Phuong 2010; Petersen *et al.* 2013); hence, conserving and managing the puerulus fishery in Palawan could help improve the economic development of the Philippines.

CONCLUSION

The attractive price per piece of puerulus during the surge of the COVID-19 pandemic [see Mecha *et al.* (2023)] could be the main reason for its rapid expansion in Palawan. In 2019, the collection started in Sofronio Española and immediately spread northward the following year, which became an important source of livelihood for those living along the eastern coast. The high concentration of collection sites in eastern municipalities such as Roxas, Narra, Sofronio Española, and Brooke's Point signify these municipalities as hotspots due to the geographical location and oceanographic advantages facing the Sulu Sea. These localities generally characterized by sandy-muddy-substrate require management interventions (including the nearby ecosystems) to sustain this invaluable spiny lobster fishery.

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STATEMENT ON CONFLICT OF INTEREST

The authors have stated that there are no conflicts of interest associated with this publication, and there has been no significant financial support for this work that could have influenced its outcome.

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